A Logistic Regression Model for Horseshoe crab data set

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1 Dataset Description

This data set is derived from Agresti (the 3rd edition, 2013, Table 4.3). In this project, we will refer to the horseshoe crab data. Some female crabs attract many males while others do not attract any. The males that cluster around a female are called satellites. In order to understand what influences the presence of satellite crabs, researchers selected female crabs and collected data on the following characteristics:

- The color of her shell:(1 - light medium, 2 - medium, 3 - dark medium, 4 - dark)
- The condition of her spine:(1 - both good, 2 - one worn or broken, 3 - both worn or broken)
- The width of her carapace shell (in centimeters).
- The number of male satellites.
- The weight of the female (in kilogram (kg)).
- Y: $Y_i = 1$ if a female crab has one or more satellites, and $Y_i = 0$ if not.

```
> str(Crabs_Data)
'data.frame':   173 obs. of  6 variables:
 $ satell: int  8 0 9 0 4 0 0 0 0 0 ...
 $ y     : int  1 0 1 0 1 0 0 0 0 0 ...
 $ weight: num  3.05 1.55 2.3 2.1 2.6 2.1 2.35 1.9 1.95 2.15 ...
 $ width : num  28.3 22.5 26 24.8 26 23.8 26.5 24.7 23.7 25.6 ...
 $ color : int  2 3 1 3 3 2 1 3 2 3 ...
 $ spine : int  3 3 1 3 3 3 1 2 1 3 ...
```

```
> head(Crabs_Data)
  satell y weight width color spine
1      8 1   3.05  28.3     2     3
2      0 0   1.55  22.5     3     3
3      9 1   2.30  26.0     1     1
4      0 0   2.10  24.8     3     3
5      4 1   2.60  26.0     3     3
6      0 0   2.10  23.8     2     3
```

In this project, we fit a logistic regression model in R with quantitative and/or categorical predictors to the Horseshoe crab data set. After evaluating the model, we introduce the optimal model using Stepwise Selection method. In the following, we calculate confusion matrix and odds ratio for the best Model and interpret the results.

2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is an analysis approach that identies general patterns in the data. These patterns include features of the data and correlation between each independent variable (predictor) and the dependent variable. EDA is an important rst step in any data analysis.

```
> summary(Crabs_Data)
     satell             y              weight          width
 Min.   : 0.000   Min.   :0.0000   Min.   :1.200   Min.   :21.0
 1st Qu.: 0.000   1st Qu.:0.0000   1st Qu.:2.000   1st Qu.:24.9
 Median : 2.000   Median :1.0000   Median :2.350   Median :26.1
 Mean   : 2.919   Mean   :0.6416   Mean   :2.437   Mean   :26.3
 3rd Qu.: 5.000   3rd Qu.:1.0000   3rd Qu.:2.850   3rd Qu.:27.7
 Max.   :15.000   Max.   :1.0000   Max.   :5.200   Max.   :33.5
     color           spine
 Min.   :1.000   Min.   :1.000
 1st Qu.:2.000   1st Qu.:2.000
 Median :2.000   Median :3.000
 Mean   :2.439   Mean   :2.486
 3rd Qu.:3.000   3rd Qu.:3.000
 Max.   :4.000   Max.   :3.000
```

```
Correlation Matrix
> corr_mat
            y weight  width  color  spine
y       1.000  0.387  0.401 -0.268 -0.028
weight      .  1.000  0.887 -0.251 -0.166
width       .      .  1.000 -0.264 -0.122
color       .      .      .  1.000  0.379
spine       .      .      .      .  1.000
```

from correlation matrix, we observe that the highest correlation value is between **weight** and **width** variables.

3 Logistic Regression Model

Next, we t a logistic regression model to the data, and the results are as follows.

```
> summary(fit_logit)
Call:
glm(formula = y ~ (weight + width + color + spine)^2, family = binomial(link = logit),
    data = Crabs_Data)
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.1976  -0.9736   0.4704   0.8600   2.0020
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  16.63968   20.15214   0.826    0.409
weight       -4.42607    6.39019  -0.693    0.489
width        -0.74427    0.89195  -0.834    0.404
```

```
color        -2.20352    5.19634  -0.424    0.672
spine        -2.63901    5.33202  -0.495    0.621
weight:width  0.23296    0.21706   1.073    0.283
weight:color  0.32644    0.99820   0.327    0.744
weight:spine -0.77299    0.95257  -0.811    0.417
width:color   0.03328    0.26358   0.126    0.900
width:spine   0.18384    0.26231   0.701    0.483
color:spine  -0.01125    0.35239  -0.032    0.975
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 225.76  on 172  degrees of freedom
Residual deviance: 184.18  on 162  degrees of freedom
AIC: 206.18
Number of Fisher Scoring iterations: 6
```

3.1 Outliers & Unusual Points

Regression models are very sensitive to be affected by outliers. In a logistic model, observations whose values diverge from their expected intervals and result in a model peculiarity may be outliers. Pearson and deviance residuals are useful in identifying observations that are not explained well by the model. We will use standardized residuals to detect outliers, it is suggested the range of a standardized residual to be $(-3, +3)$, if a standardized residual is out of this range, it is an outlier, then we remove it and check the next until there is no observation with a significantly large standardized residual. Find the Table 1, this circulation resulted in total 3 observations were removed. We shall also check performance of the model based on this clean dataset again. Check the Table 2, the model has better performances in Deviance and AIC.

Table1: Outliers and Unusual Points based on Studentized Residuals.

Id	|Studentized Residuals|	y	weight	width	color	spine
87	3.201172	0	3.05	28.5	2	3
94	3.135650	0	3.20	28.7	2	1
157	3.151371	0	3.20	28.3	2	3

Table2: Performance of Models without vs. with Outliers.

Obs	Status	Deviance(P-value)	AIC
1	Data without Outliers	0.3176	188.92
2	Data with Outliers	0.1118	206.18

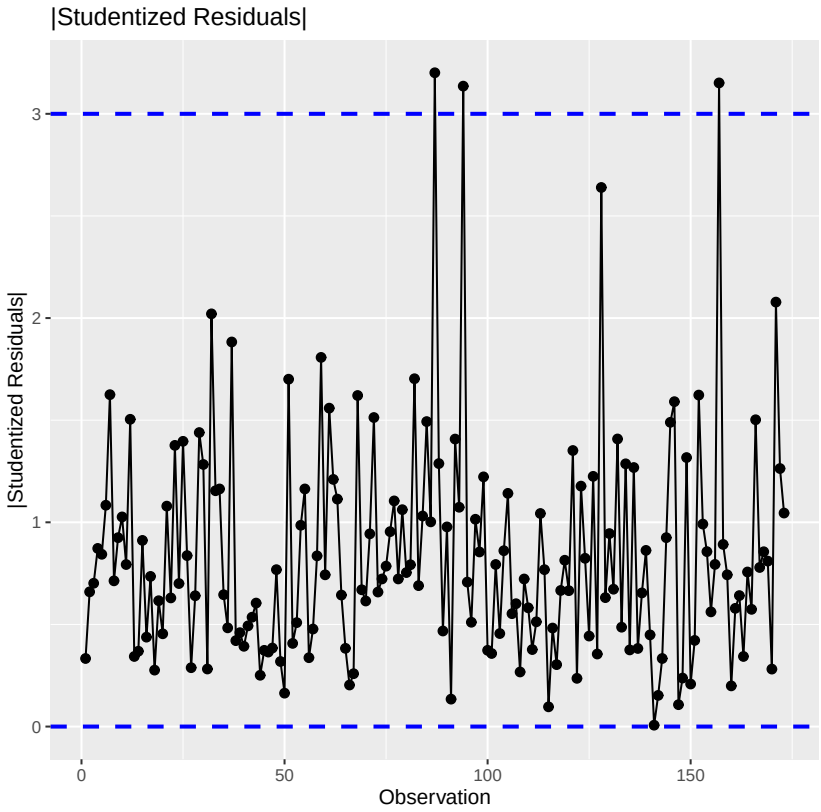


Figure1: Outliers and Unusual Points based on Studentized Residuals.

3.2 Model and Variable Selection

One of the most critical statistical issues is the problem of selecting the best model among regression models or choosing important variables in a regression model with a large number of predictor variables. Model selection is the process of choosing a model from a set of candidate models. Variable selection means choosing from among many variables to be included in a particular model, i.e., selecting relevant variables from a complete list of variables by excluding irrelevant or redundant variables. The goal of such selection is to determine a set of variables that provides the best fit for the model to make accurate predictions. In this section, to select the best model or important variables, we use Stepwise Selection method, and choose a model that has smallest values Information Akaike Criterion (AIC) .

```
> summary(best_model)
Call:
glm(formula = y ~ weight + width + color + weight:width, family = binomial(link = logit),
    data = Crabs_Data_new)
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.3294  -0.9312   0.3047   0.9265   1.6837
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   18.8590    14.8748   1.268   0.2049
weight       -11.7886     7.0370  -1.675   0.0939 .
width         -0.7381     0.5849  -1.262   0.2070
color         -0.5599     0.2347  -2.385   0.0171 *
weight:width   0.4892     0.2675   1.829   0.0675 .
---
```

```
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 219.51  on 169  degrees of freedom
Residual deviance: 169.42  on 165  degrees of freedom
AIC: 179.42
Number of Fisher Scoring iterations: 6
```

3.3 Confusion matrix and Odds ratio for the best Model

In this section, we compute the confusion matrix and odds ratio for the logistic regression model.

Table3: Odds ratio for the best Model.

Predictor Vaiables	Odds ratio(OR)
Weight	0
Width	0.48
Color	0.57
Weight:Width	1.63

- The odds ratio of 0 for weight variable implies that a 1 unit increase in "weight", the odds of that the female horseshoe crab has one or more satellites does not change.
- The odds ratio of 0.48 for width variable implies that a 1 unit increase in "width" , the odds of that the female horseshoe crab has one or more satellites decreases by a factor of 0.48.
- The odds ratio of 0.57 for color variable implies that a 1 unit increase in "color" , the odds of that the female horseshoe crab has one or more satellites decreases by a factor of 0.57.
- The odds ratio of 1.63 for the interaction between weight and width variables implies that a 1 unit increase in weight and width variables , the odds of that the female horseshoe crab has one or more satellites increases by a factor of 1.63.

Table4: Confusion Matrix.

Predicted Status	Observed Status	
	y=0	y=1
y=0	14(0.2745)	5(0.0980)
y=1	6(0.1176)	26(0.5098)

Observation from the confusion matrix the information that we can get is that, with total of 20 (y=0, sampled female has no satellites) the model has predicted 14 correctly and 6 wrongly. Similarly, total of 31 (y=1,sampled female has at least 1 satellite) the model has predicted 26 correctly and 5 wrongly. Percentage of correct predictions is $\frac{(14+26)}{(14+6+26+5)} \times 100 = 78.43\%$, which tells us 78.43% of the responses are predicted correctly.

Appendix

R code:

```
suppressWarnings({library(MASS)})
suppressWarnings({library(ggplot2)})
suppressWarnings({library(dplyr)})
suppressWarnings({library(Matrix)})
#---------------------------------------------------------------------
#                        Section 1
#---------------------------------------------------------------------
Crabs_Data<-read.table(file="D:/Horseshoe_crab_data2.txt", header=TRUE)
Crabs_Data<-Crabs_Data[,-1]
colnames(Crabs_Data)<-c("satell","y","weight","width","color","spine")
str(Crabs_Data)
head(Crabs_Data)
#---------------------------------------------------------------------
#                        Section 2
#---------------------------------------------------------------------
summary(Crabs_Data)
corr_mat<-cor(Crabs_Data[,-1],method="pearson")
corr_mat<-round(corr_mat,3)
corr_mat<-triu(corr_mat)
cat("\n Correlation Matrix\n")
corr_mat
#---------------------------------------------------------------------
#                        Section 3
#---------------------------------------------------------------------
fit_logit<-glm(y~(weight+width+color+spine)^2,
              family=binomial(link=logit),data=Crabs_Data)
summary(fit_logit)
#---------------------------------------------------------------------
#                        Section 3-1
#---------------------------------------------------------------------
aplha<-0.05
r_Stud_Res<-stdres(fit_logit)
t_range<-3
Data_outliers_Stud_Res<-abs(r_Stud_Res)[abs(r_Stud_Res)>t_range]
id_Data_outliers_Stud_Res<-which(abs(r_Stud_Res)>t_range)
result_Data_outliers_Stud_Res<-data.frame(id_Data_outliers_Stud_Res,
                                Data_outliers_Stud_Res,
                                Crabs_Data[id_Data_outliers_Stud_Res,-1])
colnames(result_Data_outliers_Stud_Res)<-c("Id","|Studentized Residuals|",
             "y","weight","width","color","spine")

data_outliers_plot<-ggplot(data.frame(1:nrow(Crabs_Data),abs(r_Stud_Res)),
           aes(x=1:nrow(Crabs_Data),y=abs(r_Stud_Res)))+
    geom_point(size=2)+geom_path()+
    geom_hline(yintercept=0,linetype="dashed",color="blue",size=1)+
    geom_hline(yintercept=3,linetype="dashed",color="blue",size=1)+
    xlab("Observation")+ylab("|Studentized Residuals|")+
    labs(title="|Studentized Residuals|")
data_outliers_plot

if(length(Data_outliers_Stud_Res)>0){
cat("\n Outliers and Unusual Points based on Studentized Residuals(r_Stud_Res)\n")
print(result_Data_outliers_Stud_Res)
}else{
cat("\n No outliers in data based on Studentized Residuals(r_Stud_Res)\n")
}
#-------------------------------------------------------------
```

```
Crabs_Data_new<-Crabs_Data[-id_Data_outliers_Stud_Res,]
#-------------------------------
Deviance1<-fit_logit$dev
P_value1<-pchisq(184.1818,162,lower.tail=T)
Deviance1
P_value1
#-------------------------------
fit_logit_new=glm(y~(weight+width+color+spine)^2,
                    family=binomial(link=logit),data=Crabs_Data_new)
summary(fit_logit_new)

Deviance2<-fit_logit_new$dev
P_value2<-pchisq(166.9232,159,lower.tail=T)
Deviance2
P_value2
#-----------------------------------------------------------------------
#                        Section 3-2
#-----------------------------------------------------------------------
best_model<-step(fit_logit_new)
summary(best_model)

nullModel<-glm(y~1,family=binomial(link=logit),data=Crabs_Data_new)
fullModel<-glm(y~(weight+width+color+spine)^2,
                     family=binomial(link=logit),data=Crabs_Data_new)
summary(stepAIC(fullModel,direction="backward",
                scope=list(upper=fullModel,lower=nullModel),
                trace=F))
#-----------------------------------------------------------------------
#                        Section 3-3
#-----------------------------------------------------------------------

odds_ratio<-exp(coef(best_model))
round(odds_ratio,2)
#-------------------------------------------------------
set.seed(1388)
Id<-1:170
Crabs_Data_new2<-data.frame(Id,Crabs_Data_new)
split<-sample.split(Crabs_Data_new2$Id,SplitRatio=0.7)

Crabs_Data_new3<-data.frame(Crabs_Data_new,split)
Crabs_Data_train<-Crabs_Data_new%>%dplyr::filter(Crabs_Data_new3$split=="TRUE")
Crabs_Data_test<-Crabs_Data_new%>%dplyr::filter(Crabs_Data_new3$split=="FALSE")

fit_logit_train<-glm(y~weight+width+color+I(weight*width),
              family=binomial(link=logit),data=Crabs_Data_train)

models.probs_test<-predict(fit_logit_train,Crabs_Data_test,type="response")
models.pred_test<-rep(0,length(models.probs_test))
models.pred_test[models.probs_test>0.5] <- 1
Table_model_performance2<-table(models.pred_test,Crabs_Data_test$y,
                          dnn = c("Predicted Status","Observed Status"))
Table_model_performance2
prop.table(Table_model_performance2)
```

References

[1] Agresti, A. (2013) Categorical Data Analysis. 3rd Edition, John Wiley Sons Inc., Hoboken.

[2] Kutner, M.H., Nachtsheim, C.J., Neter, J. and Li, W. (2005) Applied Linear Statistical Models. 5th Edition, McGraw-Hill, Irwin, New York.